

Introduction

Plastic production has sharply increased over the last 70 years. In 1950, the world produced just **two million tons**. It now produces over 450 million tons. However, when plastic waste is mismanaged. One to two million tons of plastic enter our oceans yearly, **affecting wildlife and ecosystems** (Ritchie 2023). According to the latest report by the Organization for Economic Cooperation and Development (OECD), only 9% of plastic generated is properly recycled, 19% incinerated **49% landfilled** and about 22% is mismanaged and becomes uncollected litter (OECD 2022).

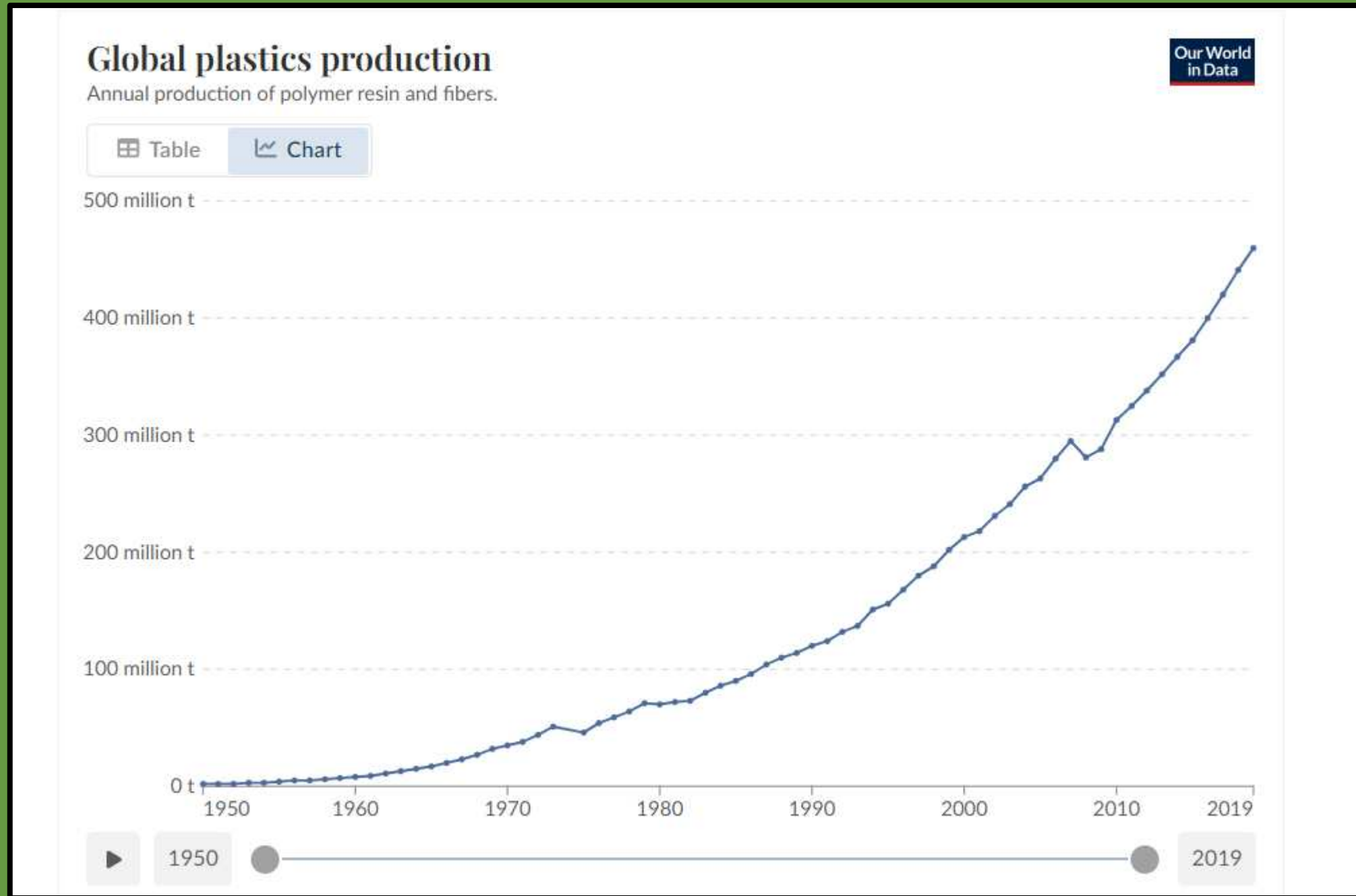
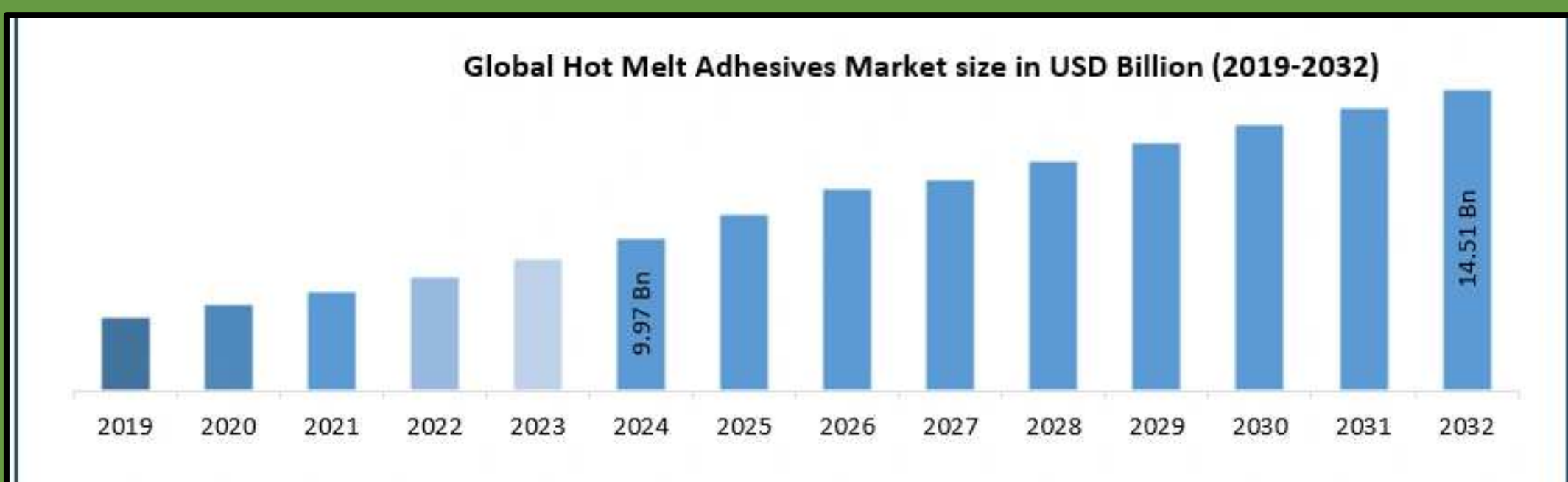


Figure 1. Global plastics production (441 Million tons at 2019) (Ritchie & Roser, 2018)

Once plastic reaches the environment in the form of macro- or microplastics, it contaminates and **accumulates in food chains** through agricultural soils, terrestrial and aquatic food chains, and the water supply. Microplastics entering the human body via direct exposures through ingestion or inhalation can lead to an array of negative health outcomes including **cancer, cardiovascular diseases, inflammatory bowel disease, diabetes, etc.** (CIEL 2025).



**Figure 2. Projected HMA Market
(Maximize Market Research Pvt. Ltd., 2024)**

Purpose

A formula to **upcycle waste PLA** into Thermoplastic Adhesives can be attempted using Citric Acid (CA), Polyethylene Glycol (PEG), and Monobutyltin Oxide(MBTO) as catalyst. The research focuses on demonstrating that waste plastics can be turned into functioning adhesive materials with the use of **environmentally friendly materials**. Upcycling plastics into adhesives would be economically viable since the global demand for thermoplastic adhesives in industries like **packaging** and **automotives** is increasing in recent years.

Upcycling waste plastic into thermoplastic adhesives with green and affordable chemicals

Methods

Material

Lab

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    B --> C[Implementation]
    C --> D[Testing]
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Analysis

Gather and Prepare chemicals

- Polylactic acid(PLA), Polyethylene Glycol 500(PEG-500), Citric Acid(CA), and Monobutyltin Oxide(MBTO) are gathered from commercial suppliers. The PLA is cleaned, dried, and cut into small pieces to improve melting efficiency.

Adhesive Performance Testing

- The adhesive is cast into thin films and conditioned before evaluation. Adhesive performance is measured using lap-shear testing in accordance with ASTM D 1875 and ASTM D-4498 standards.

Reactive Melt Blending

- The materials are combined using reactive melt blending in a heated, stirred vessel. PLA is melted first at controlled temperature, followed by the gradual addition of PEG, citric acid, and catalyst to form a homogeneous hot-melt adhesive through esterification and transesterification reactions.`

Evaluated analytical results

- Test results are compared across formulations to evaluate the strength. Trends in performance are correlated with formulation composition and processing conditions to compare with traditional adhesives

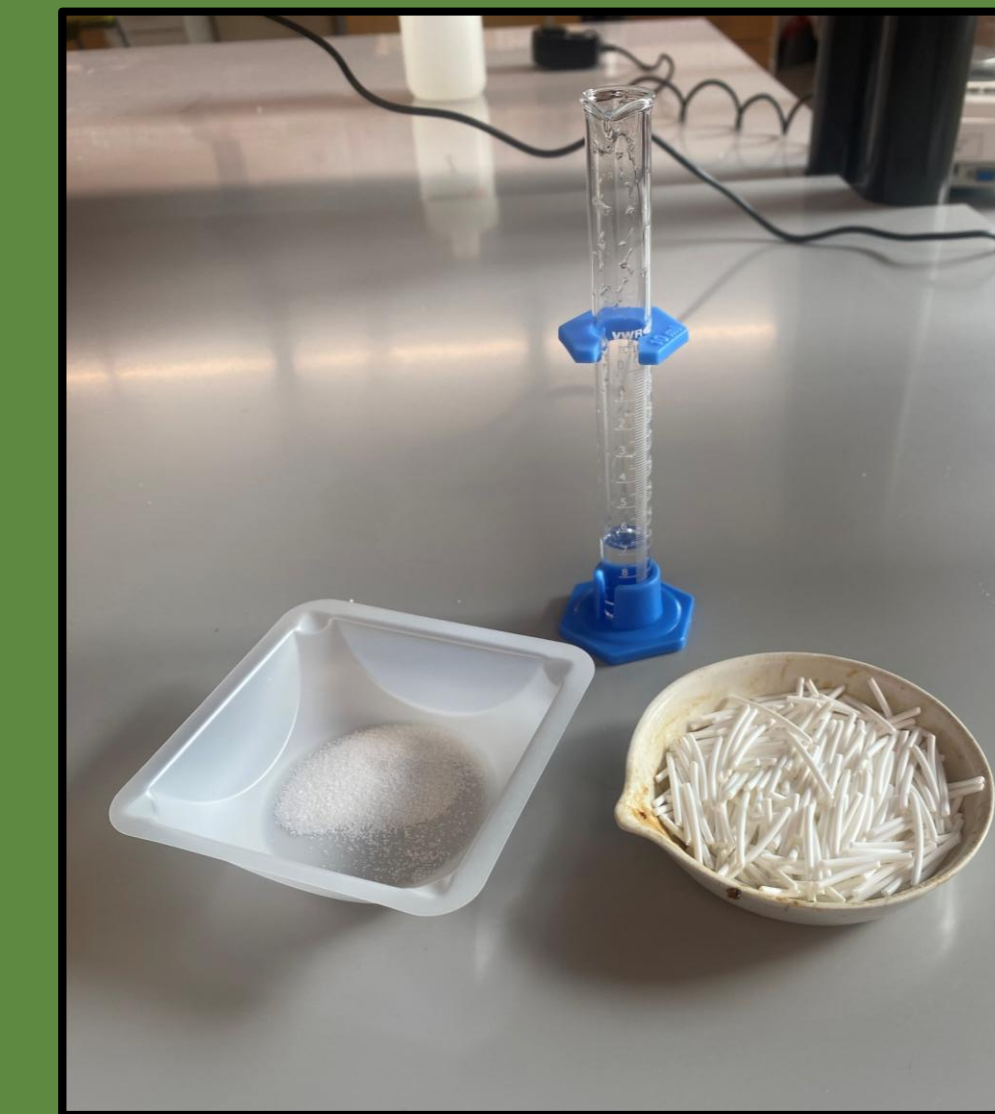


Figure 3. Materials Prepared



Figure 4. PLA half melted on hotplate



Figure 5. Tensile testing using Mark-10

Results

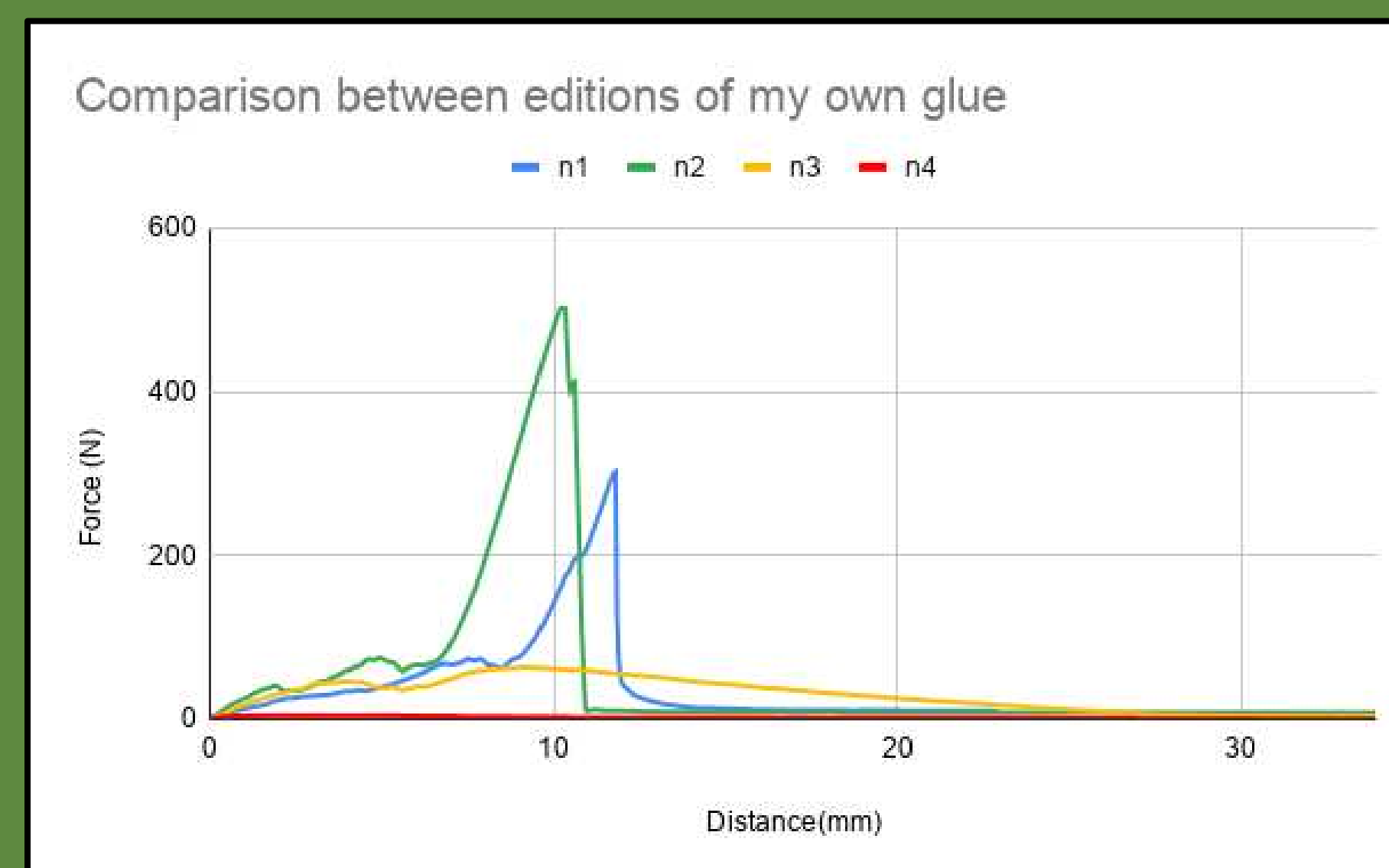


Figure 6. Strength comparison between different editions of glue.



Figure 7. From N1 to N4, there is a gradual increase of PEG, Citric Acid, and MBTO to test and modify the plastic quality of the adhesive.

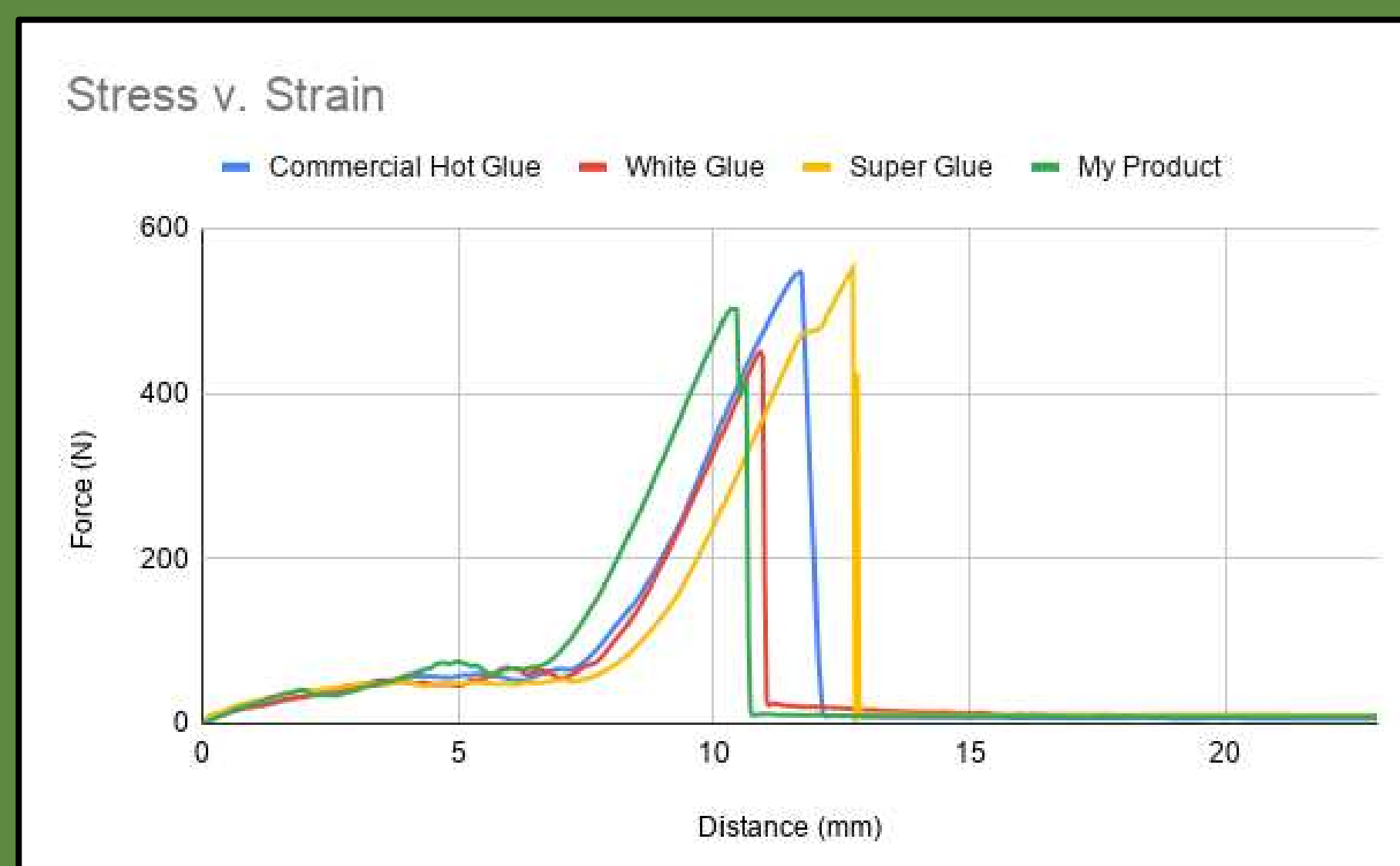


Figure 8. Strength comparison between different types of glue on the market.



Figure 9. Glue on substrate after strength testing

Conclusions

When synthesizing different editions of glue with distinct formula mixes, the name of each edition is in accordance with their date of synthesis (n1 for the earliest, n4 for the latest). The best edition, n2, is synthesized with **87.9% PLA, 9.1% PEG, 2.5% Citric Acid, and 0.4% MBTO.**

In comparison with Traditional Adhesives, the new product **performed comparably with the other test subjects** like commercial hot glue, white glue, and super glue. 502N was recorded for the n2 glue, placing it 45N below commercial hot glue and 50N below super glue.

This result was achieved using environmentally friendly chemicals. The synthesized glue demonstrated the **capability of repeatedly reheating**. The similar strength the synthesized glue yielded when compared to other glues makes of a **potential replacement** for commercial hot glue.

Extensions

The current synthesized glue was tested with a lap-shear test on a tensile tester. For this to be capable of completely replacing commercial hot glue in the **packaging and automotive industry**, it needs to be placed in more tests such as the **SAFT test** to ensure it has a stable melting temperature and strength.

Future work should evaluate the strength of the synthesized glue on **more types of substrates** such as metal and glass to mimics its working environments in the various industries. Testing could be expanded to investigate the properties of the synthesized glue or **develop a newer and better formula** with more precise control over the amount and purity of the chemicals added.



Figure 10. SAFT tester, 400C capability, self-contained, Model 2HP. Used for SAFT testing.

Citations

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